

Associations between child physical abuse potential, work-related burnout, mental health and executive functions in Italian professionals working with children and youth: A preliminary study

Alessio Porreca^{a,*}, Alessandra Simonelli^a

^a Department of Developmental and Social Psychology, University of Padova, Padova, Italy

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ABSTRACT

Background: Although commonly assumed to occur only in domestic settings with family caregivers, maltreatment is likely to happen in childcare settings too, perpetrated by professionals working with children and youth and leading to severe consequences on children's well-being. Most studies on institutional maltreatment focused on outcomes in children, without investigating the role of individual psychological variables in perpetrators, which may partly account for the occurrence of the event. For example, it is well known that professionals working with children and youth are at higher risk of experiencing work-related burnout, which in parents might account for poor parenting behaviors. Moreover, work-related burnout is associated with poor executive functioning, which may also affect the quality of caregiving. Finally, both burnout and executive function could be associated with mental health problems, which could increase the risk for maltreatment. Although the role of work-related burnout and executive functioning has been extensively studied in parents, no study has ever considered these aspects and their relationship with risk for child abuse in professionals working with children and youth.

Objective: The present study aimed to investigate risk for child physical abuse, work-related burnout, and executive functioning in Italian professionals working with children and youth aged 0–18, testing their associations with socio-demographic and work-related variables. Secondly, the study aimed to investigate the role of work-related burnout, mental health problems, and executive functioning in predicting the risk of child physical abuse in professionals working with children and youth.

Participants and setting: The study involved 125 professionals working with children and youth (79.4 % female) aged between 21 and 63 years ($M = 40.50$, $SD = 11.80$), recruited from educational and residential facilities that care for children and adolescents.

Methods: A cross-sectional study design was adopted. Participants were administered a comprehensive battery aimed at investigating socio-demographic and work-related characteristics, including presence/absence of mental health problems, risk for child abuse potential (BCAP), work-related burnout (MBI), and executive functioning (BRIEF-A).

Results: The results indicated that, according to cut-off values, 14.3 % of participants were at risk for child abuse, 20.8 % exhibited burnout symptoms, 20.8 % reported the presence of past/current mental health problems, and 5.0 % showed executive functioning difficulties. Risk for

* Corresponding author at: Department of Developmental and Social Psychology, University of Padua, Via Venezia, 8, 35131, Padua, PD, Italy.
E-mail addresses: alessio.porreca@unipd.it (A. Porreca), alessandra.simonelli@unipd.it (A. Simonelli).

child abuse was significantly associated with both work-related burnout and executive functioning. The interaction between the two was not significant, suggesting independent effects. Further analyses highlighted that the presence of mental health problems, such as anxiety and depression, was also associated with risk for child abuse.

Conclusions: The findings provide evidence for the association between work-related burnout, executive functioning, presence of mental health problems, and risk for child abuse among professionals working with children and youth. Implications for clinical practice are discussed.

1. Introduction

Childhood maltreatment represents a global public health and social welfare concern, affecting the lives of millions of children all over the world (Finkelhor et al., 2015; Stoltenborgh et al., 2015), and refers to any interaction/lack of interaction, reasonably under a caregiver's control, that does potential harm to children aged 0–18 (Brodie & Pearce, 2017; van IJzendoorn et al., 2020).

Child maltreatment affects over 55 million children in Europe (Sethi et al., 2013), with severe consequences on the short- and long-term health of involved individuals (Hughes et al., 2017), finally resulting in substantial and avoidable health-care costs throughout the life-span (Hughes et al., 2017; Sethi et al., 2018). Child physical abuse represents one of the most frequent forms of maltreatment that occurs and remains often underreported or undetected (Kellogg et al., 2007). Nearly 3 in 4 children in toddlerhood or in the preschool period regularly experience physical punishment or psychological violence at the hands of parents and caregivers every year (UNICEF, 2020). Experiencing physical abuse in the first five years of life predicts worse outcomes in physical health, mental well-being, and criminal behaviors (Lansford et al., 2021). Notably, research has shown that a history of maltreatment experienced during childhood could increase the risk of perpetrating maltreatment later in life, suggesting the presence of an intergenerational transmission (Alink et al., 2019).

1.1. Child maltreatment in institutional settings

Although most forms of childhood maltreatment are believed to occur in domestic settings at the hands of family caregivers, there is evidence that such episodes might happen in institutional settings too (Faller, 1988; Finkelhor et al., 2015; Schumacher & Carlson, 1999; Talmon et al., 2024). Some authors refer to institutional maltreatment to indicate the abusive and/or neglectful acts perpetrated by teachers, directors, non-professionals or volunteers, family members of staff, and peers in contexts such as schools, out-of-home care, youth/juvenile detention, sport clubs, recreational settings, religious organizations, or other comparable child and youth serving organizations in which children live or spend time (Attar-Schwartz, 2011; Brodie & Pearce, 2017; Finch et al., 2021; Finkelhor et al., 2015; Kamavarapu et al., 2017; Rus et al., 2017; Talmon et al., 2024). For the purpose of this work, we will refer to child maltreatment occurring in institutions in a broad perspective, including both single episodes of maltreatment carried out by professionals and forms of maltreatment that can be more structured, reiterated, and supported by the organizational culture (Gil, 1982).

In several of these contexts, as educational settings (e.g., daycare centers and schools) or residential facilities, professionals often struggle to care for large numbers of children and their complex needs and may be poorly trained, underpaid, and lack motivation (Gallant, 2009; Juffer et al., 2017; Kostantopoulou & Mantziou, 2020). Furthermore, professionals in residential facilities are not always prepared for the challenging behaviors of traumatized and abused children and youth, and lack information about their past history (Kostantopoulou & Mantziou, 2020). Given that childcare providers become actual attachment figures for children (Goossens & van IJzendoorn, 1990), institutional maltreatment can be considered a form of relational trauma (Talmon et al., 2024).

Institutional maltreatment is related to a series of clinical outcomes for the abused individual, in line with consequences experienced in relation to abuse occurring in family context (Kelley et al., 1993; Talmon et al., 2024). Significant differences that could lead to diverse outcomes exist between diverse care settings and could be related to the type of abuse, age of the children, reasons for their placement, and characteristics of the staff. For example, young children who experienced sexual abuse in daycare are more likely to exhibit problems concerning emotions, behaviors, toilet training, and development (Vrolijk-Bosschaert et al., 2017) whereas preschool children experiencing verbal abuse by teachers during kindergarten are at higher risk of developing delinquent behavior and academic difficulties in early adolescence, although this effect could be partly related to individual characteristics (Brendgen et al., 2006). Similar results were also found in school-aged children and adolescents who were victims of sexual abuse in schools, presenting several psychological problems, as high emotional distress, somatic symptoms, and behavioral problems (Gomes-Schwartz et al., 1985). Notably, many of these outcomes are likely to persist even several years after the event (Burgess & Hartman, 2005).

Regarding long-term care institutions, studies focusing on residential care highlighted the presence of several forms of maltreatment occurring at different ages and linked to severe behavioral or emotional problems in over half of the children (Hobbs et al., 1999). Moreover, the prevalence of psychological disorders among adult survivors of institutional abuse in industrial schools and reformatories was over 80 % and far higher than in the general population, with anxiety, mood, and substance use disorders being the most prevalent diagnoses, followed by high rates of trauma symptoms and insecure adult attachment styles (Carr et al., 2010). Together, these results highlight the presence of significant associations between the experience of abuse in settings that provide care to children and youth and adjustment across the life span in the domains of mental health, physical well-being, and psychosocial adjustment (Carr et al., 2020).

It is often taken for granted that personnel in these contexts are extensively trained and able to provide adequate

caregiving to children, enhancing the possibility that maltreatment in settings that provide care to children and youth receives less attention and remains a phenomenon more difficult to detect, with the risk that outcomes are misinterpreted and that victims are not provided proper and timely interventions (Talmon et al., 2024). Notably, in a systematic review of the international literature on maltreatment in large institutional settings, such as orphanages in developing and developed countries, rates of maltreatment, including physical and sexual abuse, ranged from 13 % to 93 % (Sherr et al., 2017).

1.2. Variables associated with child abuse

Most of the research on institutional abuse focused on the outcomes of children and youth; less is known about the antecedents that put childcare workers at risk of perpetrating child maltreatment. Some studies considered the role of work-related characteristics, indicating that structural features (e.g., adult/child ratios, group sizes, and staff characteristics) and process variables (e.g., interactions between adults and children) might increase workload and the risk of turnover (Schumacher & Carlson, 1999; Whitebook, 1989). Centers with systemic and structural difficulties, such as job turnover, can be characterized by classrooms with less developmentally appropriate environments and activities, and teaching staff in these programs are likely to interact less sensitively and appropriately with children (Phillips et al., 1991). Less is known about the role of individual psychological variables that may partly account for the occurrence of the event.

For example, it is well known that professionals working with children and youth are at higher risk of experiencing work-related burnout, which is predictive of lower productivity and effectiveness (Maslach et al., 2001), higher turnover (Ducharme et al., 2008), work-related mental health problems (Corr et al., 2015), such as sleep disorders (Armon et al., 2008) and alcohol abuse (Ahola et al., 2006), and finally lower quality of care (Løvsgren, 2016). It has been suggested that work-related burnout could be linked both to individual (e.g., control and reward) and organizational (e.g., workload) variables (Blöchliger & Bauer, 2018). Notably, in parents of toddlers and preschoolers, work-related stress, a precursor of burnout, has been linked to less optimal parent-child interactions, characterized by withdrawal, expressions of anger, and a greater use of harsh discipline in both mothers (Repetti & Wood, 1997) and fathers (Repetti, 1994). Moreover, work stress was predictive of mother- and father-offspring tensions also during adolescence (Almeida & McDonald, 1998), suggesting the possibility that work-related strains might affect quality of care and risk of child maltreatment at different stages of development. Finally, work-related burnout has been found to significantly impact quality of family life, being associated with displays of anger, absence from and lack of involvement in care (Jackson & Maslach, 1982). The literature indicates that chronic stress and burnout are risk factors for the development of mental health problems, such as depression and anxiety among social workers (Maddock, 2024). For example, epidemiologic studies highlight that the risk of depressive disorders tends to be greater when burnout is severe (Ahola et al., 2005). Remarkably, the presence of mental health problems has been linked to less optimal parent-child interactions (Sethna et al., 2015; Stevenson-Hinde et al., 2013) and to an increased risk of child maltreatment in parents, both in the perinatal period (Ayers et al., 2019; Chandra et al., 2006) and later on (Chemtob et al., 2013).

Alongside this, burnout has been linked to difficulties in cognitive functioning, with professionals reporting high levels of emotional exhaustion showing more errors and longer reaction times on tasks involving executive functioning (Diestel et al., 2013). These results were also supported by recent meta-analytic work highlighting that clinical burnout is associated with broad impairments across several cognitive domains, including executive functions, with effect sizes ranging from small to medium (Gavelin et al., 2021). Moreover, research showed that poor executive functioning mediates the negative relationship between burnout and task performance at work (Lemonaki et al., 2021). Notably, cognitive functioning has also been linked to mental health problems, with higher dysfunctions associated with an increase in psychopathological symptoms (Chavez-Baldini et al., 2023). Moreover, difficulties in executive functions have been extensively linked to dysfunctional caregiving in parents (Deater-Deckard et al., 2012a; Park & Johnston, 2020), suggesting a potential role in linking distal factors of child maltreatment to caregiving practices and subsequent child outcomes (Gonzalez, 2015). For example, deficits in working memory were associated with greater negative and reactive parenting behaviors, and parents with lower performance on executive functioning were more likely to engage in harsh parenting practices (Deater-Deckard et al., 2012b; Deater-Deckard & Kirby, 2017). Moreover, additional findings highlighted that parental inhibitory control was related to both sensitive and intrusive parental behaviors (Musser et al., 2012; Shaffer & Obradović, 2017).

As far as we know, no study has considered the role of work-related burnout and executive functioning in professionals working with children and youth. These aspects could be of particular importance given that emotional exhaustion resulting from burnout and impairments in executive functioning could predispose personnel to low-quality caregiving practices (e.g., verbal or physical intrusiveness, reactivity, harsh disciplinary strategies) and, subsequently, to a higher risk of child maltreatment.

1.3. Child abuse in Italy

The most recent report highlights that in 2018 45/1000 children were taken into care by social services in Italy, and 19.3 % of these children suffered some form of maltreatment (Martinelli et al., 2021). Physical abuse accounted for 9.6 % of the instances, with an increase from 8.6 % to 11.9 % between 2013 and 2018, representing 70 % of case filings (Martinelli et al., 2021). In 91.4 % of cases perpetrators were family members, whereas in 8.6 % of cases were non-family members (Martinelli et al., 2021). Regarding professional abuse, news events of the last years highlight an increasing yet undervalued phenomenon in educational institutions (Paradiso, 2018). Educational and residential services in Italy are mainly organized at a local level, presenting both a common cultural ground as well as differences in characteristics (e.g., number of children, hours of permanence in the facility, children-staff ratio) across regions and cities (Cicognani et al., 2008; James et al., 2022; Musatti & Picchio, 2010). As far as it concerns educational services, over 96 % of preschool-aged children attend kindergartens, but enrollments of infants and toddlers in nursery schools are also increasing in recent

years, due to higher rates of mothers' employment and local provision of services (Musatti & Picchio, 2010). As far as it concerns residential treatment, Italy has a comparatively low rate of children in out-of-home care – 2.7 per 1000 (Licursi et al., 2013). The majority of youths in residential care are in their teen years (close to two-thirds are between 11 and 17 years old), slightly more than one-third are ten or younger, with 12.7 % reported as younger than two years old (James et al., 2022). In Italy, children enter residential care for a wide range of reasons related to problems in parenting, relational difficulties, and family violence (James et al., 2022). Forms of maltreatment in Italian institutions range from single episodes to structured institutional violence, where maltreatment is structured across multiple organizational levels (e.g., adult-centric and rigid educational approaches, absence of supervision, omission to report cases of violence from colleagues), highlighting the need for further investigation and the promotion of a child-oriented educational culture (Paradiso, 2018). Usually, physical and psychological abuse are more frequent in nursery schools and kindergartens, whereas psychological abuse is more widespread in primary and secondary schools (Paradiso, 2018). Overall, there is a paucity of data available at regional and national levels, limiting the possibility of fully representing the phenomenon of child abuse in Italy (Cardinali et al., 2020; Martinelli et al., 2021). This is mainly due to a fragmented socio-economic framework that pairs a reported high quality of life with profound territorial differences linked to employment rates, productivity, public debt, aging population, and corruption and criminality (Cardinali et al., 2020). Moreover, most of the children (54 %) followed by social services are aged between 11 and 17, highlighting the need to identify risk in younger children (Martinelli et al., 2021).

1.4. Objectives of the study

To fill the gap in the literature, this current cross-sectional study aims to investigate risk for child maltreatment, work-related burnout, and executive functioning in professionals working with children and youth (aged 0–18) in Italy, examining their associations with socio-demographic and work-related variables (Objective 1). Secondly, the study aims to test the role of work-related burnout (Objective 2), executive functioning (Objective 3), and their interaction (Objective 4) on risk for child maltreatment, specifically child physical abuse, in this population. In line with previous studies on professionals working with children and youth (e.g., Barford & Whelton, 2010; Seti, 2008)) we expect the variables considered to present associations with diverse socio-demographic and work-related characteristics of the participants (H1). Moreover, in line with previous studies on parents (Park & Johnston, 2020; Repetti & Wood, 1997) we expect to find significant associations between work-related burnout (H2), executive functioning (H3), and risk for child physical abuse in professionals working with children and youth. Specifically, we expect professionals reporting higher levels of burnout symptoms and lower levels of executive functioning to present a higher risk for child physical abuse. Moreover, in line with studies reporting the presence of associations between workload and cognitive performance (Gavelin et al., 2021), we expect to find a significant interaction between work-related burnout and executive functioning in predicting risk for child abuse (H4). Finally, given previous studies highlighting associations between child maltreatment, work-related burnout, and executive functioning with mental health (Ayers et al., 2019; Chavez-Baldini et al., 2023; Maddock, 2024), we will also test for the effect of the latter (Objective 5), expecting a significant impact of mental health problems on risk for child abuse (H5). Specifically, we expect that individuals reporting the presence of mental health problems show a higher risk for child abuse.

Table 1
Sample characteristics.

Socio-demographic characteristics	M (SD)	N (%)
Age	40.50 (11.80)	–
Sex assigned at birth		
Male	–	25 (20)
Female	–	100 (80)
Italian Nationality	–	116 (98.3)
Education		
Middle school	–	2 (1.6)
High school diploma	–	34 (27.2)
MBA degree	–	16 (12.8)
Master's degree	–	63 (50.4)
Phd/Specialization	–	10 (8.0)
Marital status		
Single	–	46 (36.5)
Married	–	48 (38.1)
Cohabiting	–	23 (18.3)
Separated/Divorced	–	53 (42.1)
Presence of mental health problems	–	26 (20.8)
Depression	–	1 (3.8)
Anxiety	–	2 (7.7)
Depression and anxiety	–	17 (65.4)
Eating disorders	–	6 (23.1)
History of maltreatment		
Maltreatment in infancy	–	10 (7.9)
Maltreatment in adulthood	–	12 (9.5)

Note: M = mean; SD = standard deviation; N = number of subjects reporting the variable; % = percentage of the subject that reported the variable.

2. Methods

2.1. Participants

Participants were 125 professionals working with children and youth (79.4 % females) aged between 21 and 63 years old ($M = 40.50$, $SD = 11.80$) recruited from facilities working with children and adolescents (age range 0–18) in northern (63.2 %) and central (36.8 %) Italy. Sample characteristics are presented in Table 1 and in Table 2.

2.2. Study design and procedure

The study was developed through a cross-sectional design, and data collection took place from January to December 2024. Professionals working with children and youth were recruited through contact with institutions or word of mouth. Once they agreed to participate in the study, the participants signed informed consent and were administered a battery aimed at assessing different measures, including risk for child abuse, work-related burnout, and perceived executive functioning. Before completing the battery, the participants were administered a checklist investigating socio-demographic and clinical information as presence/absence of mental health problems, past life events, and data inherent job organization. The study was approved by the Ethical Committee of the University of Padova (Protocol: 4939).

2.3. Measures

2.3.1. Socio-demographic variables, mental health, and work-related variables

Variables inherent socio-demographic characteristics (Age, Sex assigned at birth, Nationality, Education, Marital status), presence of current/past mental health problems, previous experiences of maltreatment (in infancy or in adulthood) and work-related variables (Type of Institution, Work position, Length of work collaboration, Working hours per week, Number of colleagues, Number of children cared for, Weekly hours spent with children) were obtained through the administration of a structured checklist.

More specifically, as far as it concerns the variable inherent mental health, the participants were asked whether they had experienced the presence of current/past mental health problems (dichotomous answer: yes/no) and to specify which kind (multiple choice question with the possibility to provide more than one response).

2.3.2. Risk for child abuse

Risk for child abuse was measured through the Brief Child Abuse Potential Inventory (BCAP; Ondersma et al., 2005), a self-report questionnaire consisting of a 25-item Abuse Risk scale and a nine-item Validity scale comprising a six-item Lie scale and a three-item Random Responding scale. The items receive response on a dichotomous scale, agree/disagree (Item example: “Children should never disobey”). Previous studies reported that the measure was highly correlated with the longer CAP (Milner & Wimberley, 1979), already

Table 2

Work-related characteristics.

Work-related characteristics	M (SD)	N (%)
<i>Type of facility</i>		
Family community (a)	–	1 (0.8)
Educational residential care facilities for children and adolescents (b)	–	44 (34.9)
Residential care facilities for autonomy (c)	–	3 (2.4)
Therapeutic residential care facilities for children and adolescents (d)	–	14 (11.1)
Nursery school	–	2 (1.6)
Kindergarten	–	5 (4.0)
Elementary school	–	43 (34.1)
Middle school	–	14 (11.1)
<i>Work position</i>		
Psychotherapist	–	5 (3)
Nurse auxiliary	–	6 (4.8)
Educator	–	39 (31.0)
Director	–	2 (1.6)
Coordinator	–	5 (4.0)
Teacher	–	56 (44.1)
Other	–	13 (10.2)
<i>Length of work collaboration (months)</i>	126.38 (125.28)	–
<i>Working hours per week</i>	29.16 (9.06)	–
<i>Number of colleagues</i>	10.26 (10.82)	–
<i>Number of children cared for</i>	23.87 (29.35)	–
<i>Weekly hours spent with children</i>	25.60 (10.18)	–

Note: M = mean; SD = standard deviation; N = number of subjects reporting the variable; % = percentage of the subjects that reported the variable. Residential treatment in Italy is provided to children and youth who lack adequate family support. Our sample included individuals working in family-type residential care homes (a), facilities that provide social and educational care (b), facilities that provide additional psychotherapy (c), and facilities aimed at promoting personal abilities in young people close to age 18 (d).

tested in Italy (Miragoli et al., 2015), and with family characteristics associated with child maltreatment. The BCAP Abuse Risk scale presented strong internal validity in different clinical (Ondersma et al., 2005) and non-clinical samples (Walker & Davies, 2012). Moreover, the BCAP has been previously effectively used with childcare providers (Lee & Sung, 2022). For the present study, the instrument was translated and back-translated with permission from the author. The Abuse Risk scale reported good internal consistency ($\alpha = 0.829$), and in line with previous work, a score of 12 was set as a cut-off indicating risk for child abuse (Ondersma et al., 2005).

2.3.3. Work-related burnout

Work-related burnout was assessed with the Italian version of the Maslach Burnout Inventory (MBI; Maslach et al., 1997; Sirigatti & Stefanile, 1993), a self-report questionnaire consisting of 22 items divided into three subscales: emotional exhaustion (nine items), depersonalization (five items), and personal accomplishment (eight items). The items receive a response on a 7-point Likert scale, ranging from never (0) to every day (6) (Item example: "I feel emotionally drained by my work"). The instrument has previously been used in the Italian context with health care professionals (Brusaferro et al., 2000). For the present study, the Emotional exhaustion scale reported good internal consistency ($\alpha = 0.896$), whereas internal consistency was acceptable for the Personal accomplishment scale ($\alpha = 0.749$) and questionable for the Depersonalization scale ($\alpha = 0.632$), respectively. Thus, analyses were performed only on the Emotional exhaustion scale, setting cut-off scores above 1 standard deviation from the Italian validation sample (Sirigatti & Stefanile, 1993).

2.3.4. Executive functioning

Executive functioning was assessed with the Italian version of the Behavior Rating of Executive Functioning Adult Version (BRIEF-A; Gritti et al., 2017; Roth et al., 2005), a 75-item self-report questionnaire containing items aimed at assessing executive functioning, with higher scores reflecting greater difficulties experienced by the individual. The items receive a response on a 3-point Likert scale ranging from never (0) to often (2) (Item example: "I am impulsive"). The questionnaire yields an overall score, the Global Executive Composite (GEC), which is a composite of two index scores, the Behavioral Regulation Index (BRI) and the Metacognitive Index (MI). The BRI is comprised of four scales (Inhibit, Shift, Emotional Control, and Self-Monitor), and the MI is comprised of five scales (Initiate, Working Memory, Plan/Organize, Task Monitor, and Organization of Materials). Standard scores are calculated for each of the clinical scales, indices, and the summary composite. T scores are based on comparison to the normative sample, setting a clinical cut-off for scores equal to or above 65. For the present study, analyses were carried out on the Global Executive Composite, which reported excellent internal consistency ($\alpha = 0.943$).

2.4. Statistical analyses

Data were analyzed using Jamovi, version 2.2.5 (The [The jamovi project, 2025](#)). First, descriptive statistics were run to examine mean scores, frequencies, and percentages on the variables of interest. Data inherent risk for child abuse, work-related burnout, and executive functioning presented skewed distribution and were subsequently subjected to square root transformation. Differences inherent risk for child abuse, work-related burnout and executive functioning were investigated with respect to socio-demographic (age, sex assigned at birth, presence of mental health problems, history of maltreatment in infancy, history of maltreatment in adulthood), and work-related (type of facility, length of work collaboration, working hours per week, number of colleagues, number of children cared for, weekly hours spent with children) to test for H1. For this purpose, the variable type of facility was converted into a dummy variable (residential facilities vs educational and scholastic facilities). The Mann-Whitney *U* Test was adopted for categorical variables, whereas Spearman's Rank Correlation was used with continuous variables.

To test for H2, H3, H4, and H5, a General Linear Model was conducted to examine the relationship between risk for child abuse, as dependent variable (DV), and two independent variables, work-related burnout and executive functioning as independent variables (IVs), while controlling for the effect of mental health. An interaction effect between the two independent variables was also tested.

3. Results

Table 3 presents the distribution of individuals at risk for each variable.

The results highlighted that 14.3 % of the participants were at risk of child abuse. Risk for child abuse was related to the presence of mental health problems ($U = 633, p < .001, r_{pb} = 0.508$), as well as to the presence of past experiences of maltreatment in childhood ($U = 354, p = .041, r_{pb} = 0.391$) or in adulthood ($U = 303, p = .002, r_{pb} = 0.557$), with moderate to large effect sizes. No differences were found with respect to socio-demographic and work-related variables on risk for child abuse ($p > .05$).

Table 3

Distribution of individuals at risk for child abuse, burnout and difficulties in executive functioning.

	M (SD)	n at risk (%)
Risk of child abuse	6.31 (4.89)	18 (14.3)
Burnout	13.3 (11.00)	26 (20.8)
Executive functioning	46.11 (10.18)	5 (4.0)

Note: Data are given as n (%), mean (standard deviation).

As shown in the table, a considerable number of individuals presented high symptoms of work-related burnout (20.8 %), which were also linked to the presence of mental health problems ($U = 632, p < .001, r_{pb} = 0.509$), and to the occurrence of past experiences of maltreatment in adulthood ($U = 313, p = .002, r_{pb} = 0.542$), with large effect sizes. Moreover, burnout was also associated with the number of colleagues ($r = 0.253, p = .009$).

Finally, 5 % of the participants reported difficulties in executive functioning. Executive difficulties were associated with age ($r = 0.222, p = .013$) and the presence of mental health problems ($U = 850, p = .009, r_{pb} = 0.333$), with moderate effect sizes. Moreover, they were associated also with work-related characteristics as the number of children cared for ($r = 0.185, p = .040$).

No differences were found with respect to the type of facility considered ($p > .05$), with professionals working in educational and residential services presenting the same levels of risk for child abuse, burnout, and executive functioning. Overall, these results partially support H1.

Results on the GLM highlighted that both the main effects of work-related burnout [$B = 0.410, SE = 0.054, t = 4.898, p \leq 0.001, \eta^2 = 0.134$] and of executive functioning [$B = 0.192, SE = 0.111, t = 2.393, p = .018, \eta^2 = 0.032$] were significant supporting both H2 and H3 respectively. The interaction effect between the two independent variables was not significant [$B = 0.038, SE = 0.063, t = 0.394, p = .529, \eta^2 = 0.002$], indicating that the effect of work-related burnout on risk for child abuse does not depend on the level of executive functioning and thus disconfirming H4.

Finally, the variable presence of mental health problems resulted also significant in predicting risk for child abuse [$B = 0.397, SE = 0.204, t = 1.998, p = .048, \eta^2 = 0.022$], supporting H5.

The overall model was significant [$F(4, 119) = 15.025, p \leq 0.001, R^2 = 0.336, \text{Adjusted } R^2 = 0.313$]. The results of the principal effects of the IV and of the interaction between burnout and executive functioning are presented in the graphs depicted in Fig. 1.

4. Discussion

The aim of the present study was to examine work-related burnout, executive functioning, and the risk for physical abuse among Italian professionals working with children and youth and to test their relationship. In accordance with our hypotheses, all the variables presented, to some extent, associations with socio-demographic and/or work-related variables, suggesting the need to consider a wider background comprising both individual and institutional characteristics when assessing quality of care and potential risk for maltreatment in childcare workers. In line with previous work on the antecedents of maltreatment (Alink et al., 2019; Stoltenborgh et al., 2015), our results support the need to consider child abuse risk potential in the light of previous childhood maltreatment experiences, although they also suggest the importance of considering also experiences that might occur later in life. Moreover, in agreement with previous studies on the impact of institutional characteristics on professionals' wellbeing (Blöchliger & Bauer, 2018; Schumacher & Carlson, 1999), these findings consider the need to take into account aspects as the number of colleagues and child/staff ratio which may increase the workload and contribute to the development of work-related burnout and difficulties in EF. Notably, we did not find differences in work-related burnout, executive functioning, and the risk for physical abuse according to type of facility (educational vs residential settings), despite the diverse characteristics that children in each of these environments might present (Kostantopoulou & Mantziou, 2020).

In line with our hypotheses, key findings indicated that both work-related burnout and executive functioning were significant predictors of risk for child abuse, although separately. Our findings are consistent with previous research on parents, suggesting that work-related stress among caregivers is linked to negative caregiving outcomes across different ages during child development (Almeida & McDonald, 1998; Repetti, 1994; Repetti & Wood, 1997). Specifically, in line with our second hypothesis, our data suggests that also in the context of childcare work, both educational and residential, the emotional exhaustion associated with burnout may reduce caregivers' capacity to manage stress, leading to an increased risk for abusive behaviors towards children. This result is particularly salient, given the growing amount of literature highlighting that professionals working with children and youth are at increased risk for experiencing work-related burnout (Blöchliger & Bauer, 2018), which could significantly impact job performance and quality of care provided to children and adolescents (Løvgrén, 2016; Christina Maslach et al., 2001). In this sense, the possibility to prevent the emergence of excessive work-related stress through institutional support could potentially act as a protective factor both for personnels' wellbeing and for the quality of care provided (Decker et al., 2002).

In line with our third hypothesis and with prior research on parents (Deater-Deckard et al., 2012b; Park & Johnston, 2020), we found that difficulties in executive functioning were associated with a higher risk for child abuse also in professionals working with children and youth, suggesting that also in this context caregivers with subtle impairments in executive functions may struggle with decision-making, emotional regulation, impulse control, and problem-solving, which are critical for providing optimal care to children. Several studies in high-risk and low-risk contexts (e.g. Bridgett et al., 2013; Deater-Deckard et al., 2012b; Porreca et al., 2018) highlight that functional executive abilities support parents in engaging in organized, flexible and goal-directed behaviors during caregiving practices, and this seems true also for professionals. Moreover, in this specific context, the role of executive functioning might be particularly salient, given that personnel is required to be flexible enough to relate to several children and to deal with changing tasks or to pay attention to new and diverse information related both to care and to organizational requirements (Lemonaki et al., 2021). Within the complex context of childrearing institutions, EF should be considered both in terms of risk factor for quality of care, so that eventual difficulties might be timely identified, and in terms of protective factor, which could instead be supported and cultivated, reducing the risk of potentially harmful practices and finally enhancing quality of caregiving.

In contrast to our fourth hypothesis, the interaction between work-related burnout and executive functioning was not significantly associated with risk for child abuse, suggesting a possible independent effect of the two variables on child abuse risk. This result is partly in line with previous research suggesting that cognitive functioning might operate separately from other areas of mental health

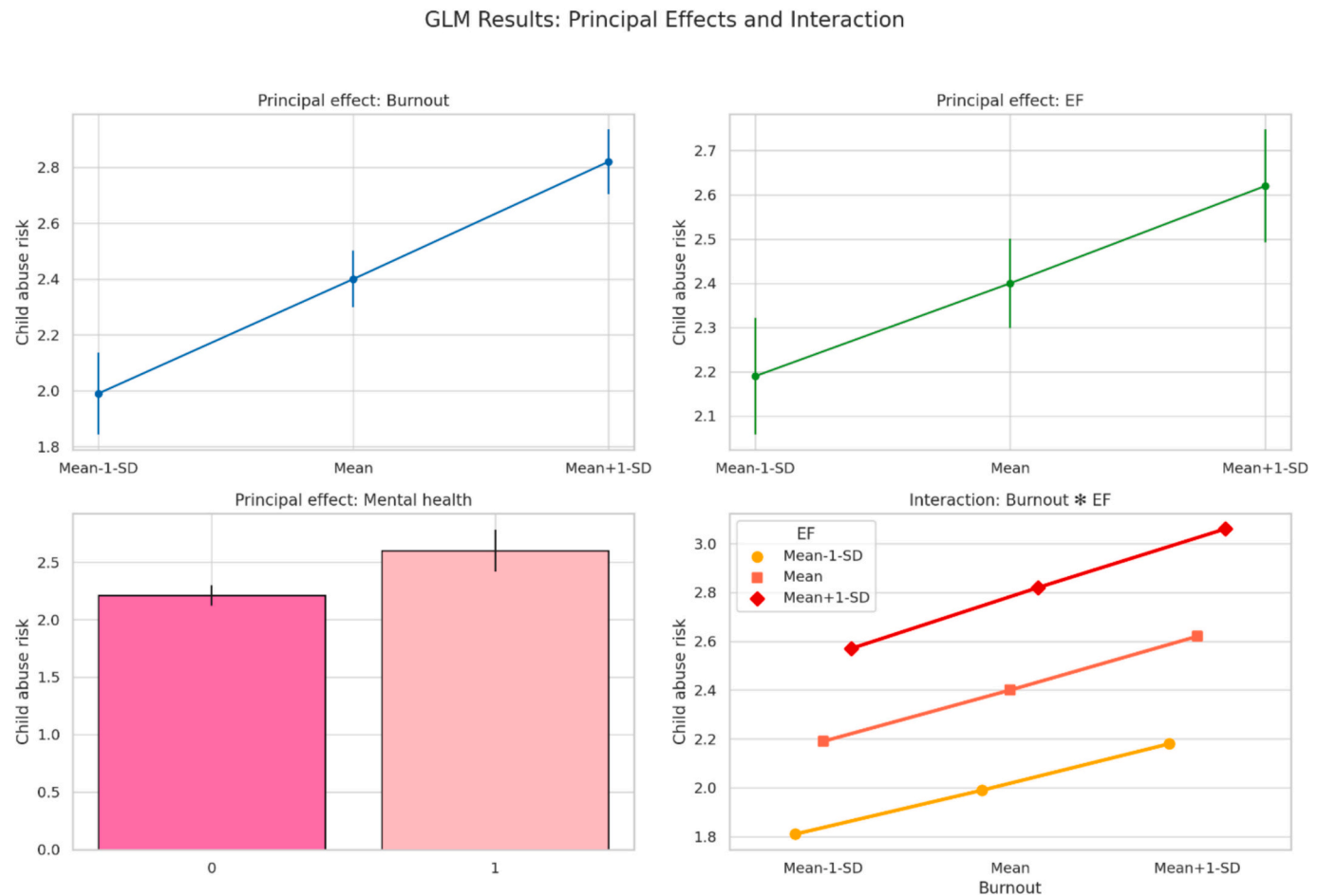


Fig. 1. Results of the GLM model.
Note. For the variable Mental health 0 = absence of past/current mental health problems, 1 = presence of past/current mental health problems. SD = standard deviation. EF = executive functioning.

in shaping quality of caregiving (Porreca et al., 2024). On the other hand, this finding could also be related to low statistical power linked to the small sample of participants in our study and should be further investigated in the future (Nosek et al., 2022; Van IJzendoorn & Bakermans-Kranenburg, 2021).

Finally, in line with our fifth hypothesis, the presence of mental health problems such as depression, anxiety, or eating disorders, also emerged as a key factor influencing the risk of child maltreatment. This aligns with previous research showing that caregivers with mental health challenges may struggle to offer children protective environments and appropriate caregiving practices (Park et al., 2006; Westad & McConnell, 2012). This aspect is of particular importance considering that almost 21 % of our sample reported the presence of current or past mental health problems, and the scarcity of studies focusing on childcare providers' mental wellbeing (Corr et al., 2014). Moreover, this aspect should be further investigated considering both the interplay between job-related stress and mental health, and in the light of the complexities that relate mental health problems, job demands, and quality of care (Corr et al., 2014; Roscoe et al., 2021).

4.1. Study limitations

The study presents a series of limitations. First it is characterized by a relatively small sample that was primarily composed of professionals working with children and youth from diverse institutional settings and just from northern and central Italy, which may reduce the generalizability of the findings. Future research should include larger and more representative samples of participants in order to better understand the associations between diverse workplaces, work-related burnout, executive functioning, and risk for child abuse. Secondly, a limitation is linked to the adoption of self-report measures, whose characteristics could further limit the generalizability of the work and point out the need to be mindful in the interpretation of our results. As far as it concerns child maltreatment, the CAP and the BCAP were in fact developed for the screening of the risk for potential abuse, implicating that the findings could include both false positives and false negatives, or could be distorted by other confounding variables as for example social desirability (Walker & Davies, 2010). As far as it concerns EF, previous work has highlighted that self-report measures might differ from battery-administrated results and might refer to more global aspects of individual functioning which could be characterized by other factors as for example personality (Buchanan, 2016). Moreover, it is to note that the presence of mental health problems was collected through a dichotomous item indicating their presence/absence, whereas having for example a measure of the current intensity of symptoms or of their lasting could have provided additional information (Haslam, 2003). Future research should thus include multi-informant assessment approaches integrating both self-report and observational measures and possibly triangulate this information with external data sources (e.g. comparing self-reported child abuse potential with actual episodes referred by the institution) to provide a more global comprehension of the phenomena of interest. Finally, a third limitation of the study is due to its cross-sectional design, which prevents the possibility to understand potential causalities between the different variables considered. Future research should include longitudinal study designs to better understand the links between work-related burnout, executive functioning, and the risk for child abuse, and to monitor their change/stability over time.

4.2. Implications for practice

Despite these limitations, the results of the present study provide a series of practical implications that could help to orient and ameliorate organizational strategies aimed at promoting childcare workers' wellbeing and at supporting quality of care provided to children.

Firstly, the findings could help to raise awareness with respect to the phenomenon of child abuse in Institutions. As the results of our work and of previous studies (Finkelhor et al., 2015; Talmon et al., 2024) highlight, even in contexts where most work tasks revolve around child protection, adults are equally at risk for enacting maladaptive caregiving practices. Thus, it is important to increase consciousness concerning maltreatment perpetrated by personnel working with children and youth among health professionals, both with respect to the self and to others, implementing the assessment of this particular form of maltreatment as part of the assessment of the general wellbeing of children and adolescents (Talmon et al., 2024). Strikingly, studies suggest that around 20 % of pediatricians and approximately one third of mental health professionals and early childhood professionals believe that spanking children aged 5 years or younger could be an effective caregiving practice, underlining the need of further knowledge and awareness on the phenomenon (Knox & Brouwer, 2008). Moreover, studies on parents highlight that the trust in corporal punishment in adults could increase the risk for child abuse (Crouch & Behl, 2001), suggesting the need, both for institutions and individuals, to monitor individual idiosyncratic beliefs concerning proper care, besides implementing assessment protocols and spreading information on the topic.

Secondly, the results of our work could help to promote organizational changes. One of the means to reduce the risk for child maltreatment could be intervening on the organization of facilities (e.g. workload adjustments inherent the number of colleagues and the number of children cared for) and on the selection, training, and supervision of personnel, promoting more favorable working aspects, as clarity relative to job expectations and a better staff/children ratio, and providing education relative to developmentally appropriate child behaviors and caregiving behaviors (Atten & Milner, 1987; Brownlee et al., 2009; Schumacher & Carlson, 1999; Talmon et al., 2024). Institutions should train new personnel to identify professionals' abuse at its earliest precursors and regularly update and supervise the services offered in order to avoid its occurrence (Kamavarapu et al., 2017). Moreover, facilities could support professionals' relational abilities in order to promote more adaptive caregiver-child interactions, and to subsequently increase quality of care (Cain et al., 2007).

Finally, the results of the present work could guide in the promotion of policies aimed at supporting staff wellbeing. As our data highlights, quality of care is not exclusively linked to organizational aspects and cannot prescind from professionals' individual health,

suggesting the need for facilities to adopt screening tools and programs (e.g. training modules, intervention protocols, and supervision sessions) aimed at supporting the staff's wellbeing. For example, protocols that address work-related burnout and executive functioning can be beneficial in preventing child abuse. Finally, identifying and addressing mental health issues among professionals working with children and youth could improve their wellbeing and finally reduce the likelihood of negative caregiving behaviors.

5. Conclusions

In conclusion, this is the first study highlighting the presence of associations between work-related burnout, executive functioning, mental health, and potential for child abuse in professionals working with children and youth. The assessment and support to professionals' work-related stressors, cognitive abilities and mental health and could reduce the risk of maltreatment, finally promoting safer environments.

CRedit authorship contribution statement

Alessio Porreca: Writing – review & editing, Writing – original draft, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Alessandra Simonelli:** Writing – review & editing, Supervision, Project administration, Funding acquisition, Conceptualization.

Declaration of competing interest

None.

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Data availability

Data will be made available on request.

References

- Ahola, K., Honkonen, T., Isometsä, E., Kalimo, R., Nykyri, E., Aromaa, A., & Lönnqvist, J. (2005). The relationship between job-related burnout and depressive disorders—Results from the Finnish Health 2000 Study. *Journal of Affective Disorders*, 88(1), 55–62. <https://doi.org/10.1016/J.JAD.2005.06.004>
- Ahola, K., Honkonen, T., Pirkola, S., Isometsä, E., Kalimo, R., Nykyri, E., Aromaa, A., & Lönnqvist, J. (2006). Alcohol dependence in relation to burnout among the Finnish working population. *Addiction*, 101(10), 1438–1443. <https://doi.org/10.1111/j.1360-0443.2006.01539.x>
- Alink, L. R. A., Cyr, C., & Madigan, S. (2019). The effect of maltreatment experiences on maltreating and dysfunctional parenting: A search for mechanisms. *Development and Psychopathology*, 31(1), 1–7. <https://doi.org/10.1017/S0954579418001517>
- Almeida, D. M., & McDonald, D. (1998). Weekly rhythms of parents' work stress, home stress, and parent-adolescent tension. *New Directions for Child and Adolescent Development*, 1998(82), 53–68. <https://doi.org/10.1002/CD.23219988205>
- Armon, G., Shirom, A., Shapira, I., & Melamed, S. (2008). On the nature of burnout-insomnia relationships: A prospective study of employed adults. *Journal of Psychosomatic Research*, 65(1), 5–12. <https://doi.org/10.1016/j.jpsychores.2008.01.012>
- Attar-Schwartz, S. (2011). Maltreatment by staff in residential care facilities: The adolescents' perspectives. *Doi:10.1086/664009. Social Service Review*, 85(4), 635–664. <https://doi.org/10.1086/664009>
- Atten, D. W., & Milner, J. S. (1987). Child abuse potential and work satisfaction in day-care employees. *Child Abuse & Neglect*, 11(1), 117–123. [https://doi.org/10.1016/0145-2134\(87\)90040-8](https://doi.org/10.1016/0145-2134(87)90040-8)
- Ayers, S., Bond, R., Webb, R., Miller, P., & Bateson, K. (2019). Perinatal mental health and risk of child maltreatment: A systematic review and meta-analysis. *Child Abuse & Neglect*, 98, Article 104172. <https://doi.org/10.1016/J.CHIABU.2019.104172>
- Barford, S. W., & Whelton, W. J. (2010). Understanding burnout in child and youth care workers. *Child and Youth Care Forum*, 39(4), 271–287. <https://doi.org/10.1007/S10566-010-9104-8/TABLES/4>
- Blöchliger, O. R., & Bauer, G. F. (2018). Correlates of burnout symptoms among child care teachers. A multilevel modeling approach. *European Early Childhood Education Research Journal*, 26(1), 7–25. <https://doi.org/10.1080/1350293X.2018.1412012>
- Brendgen, M., Wanner, B., & Vitaro, F. (2006). Verbal abuse by the teacher and child adjustment from kindergarten through grade 6. *Pediatrics*, 117(5), 1585–1598. <https://doi.org/10.1542/peds.2005-2050>
- Bridgett, D. J., Burt, N. M., Laake, L. M., & Oddi, K. B. (2013). Maternal self-regulation, relationship adjustment, and home chaos: Contributions to infant negative emotionality. *Infant Behavior and Development*, 36(4), 534–547. <https://doi.org/10.1016/J.INFBEH.2013.04.004>
- Brodie, I., & Pearce, J. (2017). Violence and alternative care: A rapid review of the evidence. *Psychology, Health & Medicine*, 22(s1), 254–265. <https://doi.org/10.1080/13548506.2017.1281980>
- Brownlee, J., Berthelsen, D., & Segaran, N. (2009). Childcare workers' and centre directors' beliefs about infant childcare quality and professional training. *Early Child Development and Care*, 179(4), 453–475. <https://doi.org/10.1080/03004430701217688>
- Brusaferro, S., Balestrieri, M., Brusaferro, S., Agnoletto, A. P., Gubian, F., & Balestrieri, M. (2000). Use of the Maslach burnout inventory to support health care workers management in hospital. Article in. *Journal of Preventive Medicine and Hygiene*, 41, 18–23. <https://www.researchgate.net/publication/255652538>
- Buchanan, T. (2016). Self-report measures of executive function problems correlate with personality, not performance-based executive function measures, in nonclinical samples. *Psychological Assessment*, 28(4), 372–385. <https://psycnet.apa.org/buy/2015-31819-001>
- Burgess, A. W., & Hartman, C. R. (2005). Children's adjustment 15 years after daycare abuse. *Journal of Forensic Nursing*, 1(2), 73–78. <https://doi.org/10.1111/J.1939-3938.2005.TB00017.X>

- Cain, D. W., Rudd, L. C., & Saxon, T. F. (2007). Effects of professional development training on joint attention engagement in low-quality childcare centers. *Early Child Development and Care*, 177(2), 159–185. <https://doi.org/10.1080/03004430500375927>
- Cardinali, P., Badalassi, G., Emmi, V., Gentile, F., & Migliorini, L. (2020). Vulnerability to child maltreatment and neglect in Italy: A proposal of an indicator. *The Open Psychology Journal*, 13(1), 193–205. <https://doi.org/10.2174/1874350102013010193>
- Carr, A., Dooley, B., Fitzpatrick, M., Flanagan, E., Flanagan-Howard, R., Tierney, K., White, M., Daly, M., & Egan, J. (2010). Adult adjustment of survivors of institutional child abuse in Ireland. *Child Abuse & Neglect*, 34(7), 477–489. <https://doi.org/10.1016/J.CHABU.2009.11.003>
- Carr, A., Duff, H., & Craddock, F. (2020). A systematic review of the outcome of child abuse in long-term care. *Trauma, Violence, & Abuse*, 21(4), 660–677. <https://doi.org/10.1177/1524838018789154>
- Chandra, P. S., Bhargavaraman, R. P., Raghunandan, V. N. G. P., & Shaligram, D. (2006). Delusions related to infant and their association with mother-infant interactions in postpartum psychotic disorders. *Archives of Women's Mental Health*, 9(5), 285–288. <https://doi.org/10.1007/S00737-006-0147-7/METRICS>
- Chavez-Baldini, U., Nieman, D. H., Keestra, A., Lok, A., Mocking, R. J. T., De Koning, P., ... Denys, D. (2023). The relationship between cognitive functioning and psychopathology in patients with psychiatric disorders: A transdiagnostic network analysis. *Psychological Medicine*, 53(2), 476–485. <https://doi.org/10.1017/S0033291721001781>
- Chemtob, C. M., Gudiño, O. G., & Laraque, D. (2013). Maternal posttraumatic stress disorder and depression in pediatric primary care: Association with child maltreatment and frequency of child exposure to traumatic events. *JAMA Pediatrics*, 167(11), 1011–1018. <https://doi.org/10.1001/JAMAPEDIATRICS.2013.2218>
- Cicognani, E., Albanesi, C., & Zani, B. (2008). The impact of residential context on adolescents' subjective well being. *Journal of Community and Applied Social Psychology*, 18(6), 558–575. <https://doi.org/10.1002/CASP.972>
- Corr, L., Davis, E., LaMontagne, A. D., Waters, E., & Steele, E. (2014). Childcare providers' mental health: A systematic review of its prevalence, determinants and relationship to care quality. *International Journal of Mental Health Promotion*, 16(4), 231–263. <https://doi.org/10.1080/14623730.2014.931067>
- Corr, L., Lamontagne, A. D., Cook, K., Waters, E., & Davis, E. (2015). Associations between Australian early childhood educators' mental health and working conditions: A cross-sectional study. *Australian Journal of Early Childhood*, 40(3), 69–78. <https://doi.org/10.1177/183693911504000310>
- Crouch, J. L., & Behl, L. E. (2001). Relationships among parental beliefs in corporal punishment, reported stress, and physical child abuse potential. *Child Abuse & Neglect*, 25(3), 413–419. [https://doi.org/10.1016/S0145-2134\(00\)00256-8](https://doi.org/10.1016/S0145-2134(00)00256-8)
- Deater-Deckard, K., & Kirby, B. (2017). Maternal executive function, heart rate, and EEG alpha reactivity interact in the prediction of harsh parenting. *Journal of Family Psychology*, 31(1), 41–50. <https://psycnet.apa.org/buy/2017-04791-004>
- Deater-Deckard, K., Wang, Z., Chen, N., & Bell, M. A. (2012a). Maternal executive function, harsh parenting, and child conduct problems. *Journal of Child Psychology and Psychiatry*, 53(10), 1084–1091. <https://doi.org/10.1111/J.1469-7610.2012.02582.X>
- Deater-Deckard, K., Wang, Z., Chen, N., & Bell, M. A. (2012b). Maternal executive function, harsh parenting, and child conduct problems. *Journal of Child Psychology and Psychiatry*, 53(10), 1084–1091. <https://doi.org/10.1111/J.1469-7610.2012.02582.X>
- Decker, J. T., Lynn Bailey, T., & Westergaard, N. (2002). Burnout among childcare workers. *Residential Treatment for Children & Youth*, 19(4), 61–77. https://doi.org/10.1300/J007V19N04_04
- Diestel, S., Cosmar, M., & Schmidt, K. H. (2013). Burnout and impaired cognitive functioning: The role of executive control in the performance of cognitive tasks. *Work & Stress*, 27(2), 164–180. <https://doi.org/10.1080/02678373.2013.790243>
- Ducharme, L. J., Knudsen, H. K., & Roman, P. M. (2008). Emotional exhaustion and turnover intention in human service occupations: The protective role of coworker support. *Sociological Spectrum*, 28(1), 81–104. <https://doi.org/10.1080/02732170701675268>
- Faller, K. C. (1988). The spectrum of sexual abuse in daycare: An exploratory study. *Journal of Family Violence*, 3(4), 283–298. <https://doi.org/10.1007/BF00989978/METRICS>
- Finch, M., Featherston, R., Chakraborty, S., Bjørndal, L., Mildon, R., Albers, B., ... Shlonsky, A. (2021). Interventions that address institutional child maltreatment: Evidence and gap map. *Campbell Systematic Reviews*, 17(1). <https://onlinelibrary.wiley.com/doi/pdf/10.1002/cl2.1139>
- Finkelhor, D., Turner, H. A., Shattuck, A., & Hamby, S. L. (2015). Prevalence of childhood exposure to violence, crime, and abuse: Results from the national survey of children's exposure to violence. *JAMA Pediatrics*, 169(8), 746–754. <https://doi.org/10.1001/JAMAPEDIATRICS.2015.0676>
- Gallant, P. (2009). Kindergarten teachers speak out: "Too much, too soon, too fast!". *Reading Horizons: A Journal of Literacy and Language Arts*, 49(3). https://scholarworks.wmich.edu/reading_horizons/vol49/iss3/3
- Gavelin, H. M., Domellöf, M. E., Åström, E., Nelson, A., Launder, N. H., Stigsdotter Neely, A., & Lampit, A. (2021). Work & stress cognitive function in clinical burnout: A systematic review and meta-analysis. *Work & Stress*, 36(1), 86–104. <https://doi.org/10.1080/02678373.2021.2002972>
- Gil, E. (1982). Institutional abuse of children in out-of-home care. *Child & Youth Services*, 4(1–2), 7–13. https://doi.org/10.1300/J024V04N01_03
- Gomes-Schwartz, B., Horowitz, J. M., & Sauzier, M. (1985). Severity of emotional distress among sexually abused preschool, school-age, and adolescent children. *Psychiatric Services*, 36(5), 503–508. <https://doi.org/10.1176/PS.36.5.503>
- Gonzalez, A. (2015). The role of maternal executive function. *Canadian Psychology*, 56(1), 46–53. <https://doi.org/10.1037/CAP0000019>
- Goossens, F. A., & van Ijzendoorn, M. H. (1990). Quality of infants' attachments to professional caregivers: Relation to infant-parent attachment and day-care characteristics. *Child Development*, 61(3), 832–837. <https://doi.org/10.1111/J.1467-8624.1990.TB02825.X>
- Gritti, E., Sgarrella, T., Leccese, A., Ginevra, M., La Malfa, G., & Soresi, S. (2017). Adattamento Italiano del BRIEF-A behavior rating inventory of executive function-adult version. Hogrefe.
- Haslam, N. (2003). Categorical versus dimensional models of mental disorder: The taxometric evidence. *Australian & New Zealand Journal of Psychiatry*, 37(6), 696–704. <https://doi.org/10.1080/J.1440-1614.2003.01258.X;PAGE=STRING:ARTICLE/CHAPTER>
- Hobbs, G. F., Hobbs, C. J., & Wynne, J. M. (1999). Abuse of children in foster and residential care. *Child Abuse & Neglect*, 23(12), 1239–1252. [https://doi.org/10.1016/S0145-2134\(99\)00096-4](https://doi.org/10.1016/S0145-2134(99)00096-4)
- Hughes, K., Bellis, M. A., Hardcastle, K. A., Sethi, D., Butchart, A., Mikton, C., ... Dunne, M. P. (2017). The effect of multiple adverse childhood experiences on health: A systematic review and meta-analysis. *The Lancet Public Health*, 2(8), e356–e366. [https://doi.org/10.1016/S2468-2667\(17\)30118-4](https://doi.org/10.1016/S2468-2667(17)30118-4)
- Jackson, S. E., & Maslach, C. (1982). After-effects of job-related stress: Families as victims. *Journal of Organizational Behavior*, 3(1), 63–77. <https://doi.org/10.1002/JOB.4030030106>
- James, S., Wilczek, L., Kilian, J., Timonen-Kallio, E., Bravo, A., del Valle, J. F., ... Rigamonti, A. (2022). A comparative analysis of residential care: A five-country multiple case-design study. *Child and Youth Care Forum*, 51(6), 1031–1062. <https://doi.org/10.1007/S10566-021-09666-6/TABLES/1>
- Juffer, F., van Ijzendoorn, H. M., & Bakermans-Kranenburg, M. J. (2017). Structural neglect in orphanages: Physical growth, cognition, and daily life of young institutionalized children in India. In *Child Maltreatment in Residential Care: History, Research, and Current Practice* (pp. 301–321). https://doi.org/10.1007/978-3-319-57990-0_14
- Kamavarapu, Y. S., Ferriter, M., Morton, S., & Völlm, B. (2017). Institutional abuse – Characteristics of victims, perpetrators and organisations: A systematic review. *European Psychiatry*, 40, 45–54. <https://doi.org/10.1016/J.EURPSY.2016.07.002>
- Kelley, S. J., Brant, R., & Waterman, J. (1993). Sexual abuse of children in day care centers. *Child Abuse & Neglect*, 17(1), 71–89. [https://doi.org/10.1016/0145-2134\(93\)90010-3](https://doi.org/10.1016/0145-2134(93)90010-3)
- Kellogg, N. D., Jenny, C., Christina, C. W., Hibbard, R. A., Spivack, B. S., & Stirling, J. (2007). Evaluation of suspected child physical abuse. *Pediatrics*, 119(6), 1232–1241. <https://doi.org/10.1542/PEDS.2007-0883>
- Knox, M., & Brouwer, J. (2008). Early childhood professionals' recommendations for spanking young children. *Journal of Child & Adolescent Trauma*, 1(4), 341–348. <https://doi.org/10.1080/19361520802505719>
- Kostantopoulou, F., & Mantziou, I. (2020). Maltreatment in residential child protection care: A review of the literature. *Dialogues in Clinical Neuroscience & Mental Health*, 3(2), 99–108. <https://www.obrela-journal.gr/index.php/obrela/article/view/171/159>
- Lansford, J. E., Godwin, J., McMahon, R. J., Crowley, M., Pettit, G. S., Bates, J. E., ... Dodge, K. A. (2021). Early physical abuse and adult outcomes. *Pediatrics*, 147(1). <https://doi.org/10.1542/PEDS.2020-0873/77104>

- Lee, J. Y., & Sung, J. (2022). Is the Brief version of the child abuse potential inventory a valid and reliable measure for childcare providers? *Early Education and Development*, 33(2), 350–368. <https://doi.org/10.1080/10409289.2021.1884467>
- Lemonaki, R., Xanthopoulou, D., Bardos, A. N., Karademas, E. C., & Simos, P. G. (2021). Burnout and job performance: A two-wave study on the mediating role of employee cognitive functioning. *European Journal of Work and Organizational Psychology*, 30(5), 692–704. <https://doi.org/10.1080/1359432X.2021.1892818>
- Licursi, S., Marcello, G., & Pascuzzi, E. (2013). Children in need in the south of Italy: Features and distortions in the deinstitutionalisation of care. *Children and Society*, 27(5), 337–349. <https://doi.org/10.1111/J.1099-0860.2011.00416.X>; JOURNAL: JOURNAL:10990860; WGROU: STRING: PUBLICATION
- Løvren, M. (2016). Emotional exhaustion in day-care workers. *European Early Childhood Education Research Journal*, 24(1), 157–167. <https://doi.org/10.1080/1350293X.2015.1120525>
- Maddock, A. (2024). The relationships between stress, burnout, mental health and well-being in social workers. *The British Journal of Social Work*, 54(2), 668–686. <https://doi.org/10.1093/BJSW/BCAD232>
- Martinelli, L., Giannotta, F., Briganti, R., Soavi, G., Devitiis, C., Crippa, A., & Patrizio, M. (2021). Il Indagine nazionale sul maltrattamento dei bambini e degli adolescenti in Italia: risultati e prospettive. <https://www.garanteinfanzia.org/sites/default/files/ii-indagine-nazionale-maltrattamento-2021.pdf>
- Maslach, C., Jackson, S. E., & Leiter, M. P. (1997). Maslach burnout inventory: Third edition. In C. P. Zalaquett, & R. J. Wood (Eds.), *Evaluating stress: A book of resources* (pp. 191–218). Sacreow Education. <https://psycnet.apa.org/record/1997-09146-011>
- Maslach, C., Schaufeli, W. B., & Leiter, M. P. (2001). Job burnout. *Annual Review of Psychology*, 52, 397–422. <https://doi.org/10.1146/ANNUREV.PSYCH.52.1.397>
- Milner, J. S., & Wimberley, R. C. (1979). Child abuse potential inventory. In *PsycTESTS dataset*. APA PsycTests. <https://doi.org/10.1037/T65940-000>
- Miragoli, S., Camisasca, E., & Di Blasio, P. (2015). Validation of the child abuse potential inventory in Italy. *SAGE Open*, 5(3). <https://doi.org/10.1177/2158244015597044>
- Musatti, T., & Picchio, M. (2010). Early education in Italy: Research and practice. *International Journal of Early Childhood*, 42(2), 141–153. <https://doi.org/10.1007/S13158-010-0011-9/TABLES/1>
- Musser, E. D., Kaiser-Laurent, H., & Ablow, J. C. (2012). The neural correlates of maternal sensitivity: An fMRI study. *Developmental Cognitive Neuroscience*, 2(4), 428–436. <https://doi.org/10.1016/J.DCN.2012.04.003>
- Nosek, B. A., Hardwicke, T. E., Moshontz, H., Allard, A., Corker, K. S., Dreher, A., ... Vazire, S. (2022). Replicability, robustness, and reproducibility in psychological science. *Annual Review of Psychology*, 73, 719–748. <https://doi.org/10.1146/ANNUREV.PSYCH-020821-114157>
- Ondersma, S. J., Chaffin, M. J., Mullins, S. M., & Lebreton, J. M. (2005). A Brief form of the child abuse potential inventory: Development and validation. *Journal of Clinical Child and Adolescent Psychology*, 34(2), 301–311. https://doi.org/10.1207/S15374424JCCP3402_9
- Paradiso, L. (2018). Child maltreatment and abuse in educational and training institutions: From the analysis of the phenomenon to the definition of prevention and intervention actions for a model of good treatment. *Formazione & Insegnamento*, XVI(1), 109–120.
- Park, J. L., & Johnston, C. (2020). The relations among stress, executive functions, and harsh parenting in mothers. *Journal of Abnormal Child Psychology*, 48(5), 619–632. <https://doi.org/10.1007/S10802-020-00622-X/TABLES/4>
- Park, J. M., Solomon, P., & Mandell, D. S. (2006). Involvement in the child welfare system among mothers with serious mental illness. *Psychiatric Services*, 57(4), 493–497. <https://doi.org/10.1176/PS.2006.57.4.493/ASSET/IMAGES/IJ10T2.JPEG>
- Phillips, D., Howes, C., & Whitebook, M. (1991). Child care as an adult work environment. *Journal of Social Issues*, 47(2), 49–70. <https://doi.org/10.1111/j.1540-4560.1991.tb00287.x>
- Porreca, A., Biringen, Z., Parolin, M., Saunders, H., Ballarotto, G., & Simonelli, A. (2018). Emotional availability, neuropsychological functioning, and psychopathology: The context of parental substance use disorder. *BioMed Research International*, 2018(1), Article 5359037. <https://doi.org/10.1155/2018/5359037>
- Porreca, A., De Carli, P., Filippi, B., Bakermans-Kranenburg, M. J., Van Ijzendoorn, M. H., & Simonelli, A. (2024). Maternal cognitive functioning and psychopathology predict quality of parent-child relationship in the context of substance use disorder: A 15-month longitudinal study. *Development and Psychopathology*, 37(1), 439–450. <https://doi.org/10.1017/S0954579424000026>
- Repetti, R. L. (1994). Short-term and long-term processes linking job stressors to father-child interaction. *Social Development*, 3(1), 1–15. <https://doi.org/10.1111/J.1467-9507.1994.TB00020.X>
- Repetti, R. L., & Wood, J. (1997). Effects of daily stress at work on mothers' interactions with preschoolers. *Journal of Family Psychology*, 11(1), 90–108. <https://doi.org/10.1037/0893-3200.11.1.90>
- Roscoe, J. N., Lery, B., & Thompson, D. (2021). Child safety decisions and parental mental health problems: A new analysis of mediating factors. *Child Abuse & Neglect*, 120, Article 105202. <https://doi.org/10.1016/J.CHIABU.2021.105202>
- Roth, R. M., Isquith, P. K., & Gioia, G. A. (2005). Behavior rating inventory of executive function - Adult version (BRIEF-A). *Archives of Clinical Neuropsychology*.
- Rus, A. V., Parris, S. R., Stativa, E., & Popa, C. O. (2017). Child maltreatment in residential centers: Summing up. In *Child maltreatment in residential care: History, research, and current practice* (pp. 509–515). https://doi.org/10.1007/978-3-319-57990-0_24
- Schumacher, R. B., & Carlson, R. S. (1999). Variables and risk factors associated with child abuse in daycare settings. *Child Abuse & Neglect*, 23(9), 891–898. [https://doi.org/10.1016/S0145-2134\(99\)00057-5](https://doi.org/10.1016/S0145-2134(99)00057-5)
- Sethi, D., Bellis, M., Hughes, K., Gilbert, R., Mitis, F., & Galea, G. (2013). European report on preventing child maltreatment. <http://www.euro.who.int/pubrequest>
- Sethi, D., Yon, Y., Parekh, N., Anderson, T., Huber, J., Rakovac, I., & Meinck, F. (2018). European status report on preventing child maltreatment. <http://www.euro.who.int/pubrequest>
- Sethna, V., Murray, L., Netsi, E., Psychogiou, L., & Ramchandani, P. G. (2015). Paternal depression in the postnatal period and early father-infant interactions. *Parenting*, 15(1), 1–8. <https://doi.org/10.1080/15295192.2015.992732>
- Seti, C. L. (2008). Causes and treatment of burnout in residential child care workers: A review of the research. *Residential Treatment For Children & Youth*, 24(3), 197–229. <https://doi.org/10.1080/08865710802111972>
- Shaffer, A., & Obradović, J. (2017). Unique contributions of emotion regulation and executive functions in predicting the quality of parent-child interaction behaviors. *Journal of Family Psychology*, 31(2), 150–159. <https://doi.org/10.1037/FAM0000269>
- Sherr, L., Roberts, K. J., & Gandhi, N. (2017). Child violence experiences in institutionalised/orphanage care. *Psychology, Health and Medicine*, 22, 31–57. <https://doi.org/10.1080/13548506.2016.1271951>
- Sirigatti, S., & Stefanile, C. (1993). MBI - Maslach burnout inventory. Adattamento e taratura per l'Italia. OS Organizzazioni Speciali. https://www.researchgate.net/publication/235656433_MBI_-_Maslach_Burnout_Inventory_Adattamento_e_taratura_per_l'Italia
- Stevenson-Hinde, J., Chicot, R., Shouldice, A., & Hinde, C. A. (2013). Maternal anxiety, maternal sensitivity, and attachment. *Attachment and Human Development*, 15(5–6), 618–636. <https://doi.org/10.1080/14616734.2013.830387>; WGROU: STRING: PUBLICATION
- Stoltenborgh, M., Bakermans-Kranenburg, M. J., Alink, L. R. A., & van Ijzendoorn, M. H. (2015). The prevalence of child maltreatment across the globe: Review of a series of meta-analyses. *Child Abuse Review*, 24(1), 37–50. <https://doi.org/10.1002/CAR.2353>
- Talmon, A., Ditzer, J., Talmon, A., & Tsur, N. (2024). Maltreatment in daycare settings: A review of empirical studies in the field. *Trauma, Violence, and Abuse*, 25(1), 512–525. https://doi.org/10.1177/15248380231155528/ASSET/IMAGES/LARGE/10.1177_15248380231155528-FIG1.JPEG
- The jamovi project. (2025). *jamovi (Version 2.6) [Computer Software]*. Retrieved from <https://www.jamovi.org>
- UNICEF. (2020). Global status report on preventing violence against children 2020. <https://srhr.dspace-express.com/server/api/core/bitstreams/139d446f-c3b2-42f2-ada1-011915a2fe37/content>
- Van Ijzendoorn, M. H., & Bakermans-Kranenburg, M. J. (2021). Replication crisis lost in translation? On translational caution and premature applications of attachment theory. *Attachment & Human Development*, 23(4), 422–437. <https://doi.org/10.1080/14616734.2021.1918453>
- van Ijzendoorn, M. H., Bakermans-Kranenburg, M. J., Coughlan, B., & Reijman, S. (2020). Annual research review: Umbrella synthesis of meta-analyses on child maltreatment antecedents and interventions: Differential susceptibility perspective on risk and resilience. *Journal of Child Psychology and Psychiatry*, 61(3), 272–290. <https://doi.org/10.1111/JCPP.13147>

- Vrolijk-Bosschaart, T. F., Brilleslijper-Kater, S. N., Widdershoven, G. A. M., Teeuw, A. H., Verlinden, E., Voskes, Y., Lindauer, R. J., Vrolijk-Bosschaart, T. F., Brilleslijper-Kater, S. N., Widdershoven, G. A. M., Teeuw, A. H., Verlinden, E., Voskes, Y., ... Lindauer, R. J. (2017). Psychosocial symptoms in very young children assessed for sexual abuse: a qualitative analysis from the ASAC study. *Child Abuse & Neglect*, 73, 8–23. <https://doi.org/10.1016/j.chiabu.2017.09.001>
- Walker, C. A., & Davies, J. (2010). A critical review of the psychometric evidence base of the child abuse potential inventory. *Journal of Family Violence*, 25(2), 215–227. <https://doi.org/10.1007/S10896-009-9285-9/TABLES/4>
- Walker, C. A., & Davies, J. (2012). A cross-cultural validation of the brief child abuse potential inventory (BCAP). *Journal of Family Violence*, 27(7), 697–705. <https://doi.org/10.1007/S10896-012-9458-9/FIGURES/1>
- Westad, C., & McConnell, D. (2012). Child welfare involvement of mothers with mental health issues. *Community Mental Health Journal*, 48(1), 29–37. <https://doi.org/10.1007/S10597-011-9374-0/TABLES/4>
- Whitebook, M. (1989). Who cares? Child care teachers and the quality of care in America. *Young Children*, 45(1), 41–45.